

A059 — SST Quantum-Number Rosetta Mega Falsifier

v0.1.2 — SST Falsifier Framework v1.0.6

1 Identification and scope

A059 is a phased, nonblocking falsification program built on the frozen E011 v0.3.0 `STATIC_READY` atlas. Individual phases are launched manually and may continue after earlier `FAIL` or `UNRESOLVED` outcomes. Such continuation is diagnostic and never upgrades an earlier result.

2 Research question

Can topology and vortex-state structure supply independent, reproducible operator candidates rather than semantic labels imposed after the fact?

3 Competing hypotheses

The null is that apparent operator structure is representation-, provider-, or definition-dependent. The alternative requires at least one independent operator candidate to survive its registered controls.

4 Assumptions and boundary

E011 is a static geometry/provenance source. It does not establish finite-core dynamics, stability, particle identity, or a field-theory charge. Missing dynamic evidence remains `UNRESOLVED`.

5 Symbols and units

All primary decision statistics in v0.1.2 are dimensionless after the E011 normalization or by construction. The new background-chirality lane is explicitly structural: it tests parity transformation rules, not a calibrated physical gravity coupling.

6 Registered equations

Machine-readable science contract

Research question Can a provenance-qualified knot/link atlas support independent, target-free topology observables with the symmetry, additivity, monodromy and closure structure required of a multi-operator particle taxonomy, without assigning semantic particle labels during blind analysis?

Objective Run a nonblocking phased falsification program: source qualification, static family structure, circulation-sign conjugation, additive signed-response discovery, frame/rotation monodromy, discrete involutions, parity-odd background chirality selection, independent second-U1 closure, and three-component/connector readiness. Each phase records its own

evidentiary boundary and later phases may run after earlier FAIL/UNRESOLVED outcomes.

H_0 H0: topology/geometry provides no reproducible operator set beyond representation choices and universal kinematic identities; any apparent cancellation, doublet, monodromy or closure is provider-sensitive, definitionally constructed, or unsupported by dynamics-ready sources.

H_1 H1: at least one preregistered operator candidate survives provider and symmetry controls while remaining independent of semantic labels, and its downstream additive/closure predictions survive held-out or source-qualified tests.

Registered equations

E1: robust static feature normalization

$$z_{jm} = \frac{x_{jm} - \text{operatorname{med}}(x_m)}{1.4826 \cdot \text{operatorname{MAD}}(x_m)} + \epsilon \quad (1)$$

Dimensional check: numerator and denominator have identical dimensions; z is dimensionless.

E2: circulation-conjugation parity

$$\text{mathbf{f}}v[-s] = -\text{mathbf{f}}v[s], \text{quad}E[-s] = E[s] \quad (2)$$

Dimensional check: each equality compares like-dimensional quantities.

E3: registered signed-response candidate

$$q_r(K, s) = s, a_r(K), \text{quad}a_r \geq 0 \quad (3)$$

Dimensional check: s is dimensionless; registered amplitudes are dimensionless.

E4: three-component 2+1 cancellation score

$$D(A, B) = \frac{\min(|2q_A + q_B|, |q_A + 2q_B|)}{3\max(|q_A|, |q_B|)} + \epsilon \quad (4)$$

Dimensional check: ratio of equal-dimension signed responses; D is dimensionless.

E5: SO(3) double-cover control

$$u(\theta) = (\cos(\theta/2), \text{hatn}\sin(\theta/2)); u(2\pi) = -1, u(4\pi) = +1 \quad (5)$$

Dimensional check: angle and quaternion components are dimensionless.

E6: frame holonomy candidate

$$h_K = \frac{1}{2\pi} \text{operatorname{Arg}}(n_0 \text{ton}_L) \quad (6)$$

Dimensional check: angle divided by angle is dimensionless.

E7: independent closure test placeholder

$$R = Q_* - T_* - \frac{1}{2}Y_* \quad (7)$$

Dimensional check: all normalized operator candidates are dimensionless.

E8: additive three-component response

$$Q_{ABC} = q_A + q_B + q_C \quad (8)$$

Dimensional check: sum of like-dimensional normalized responses.

E9: parity-odd background selector and axial branch splitting

$$\chi_H = \mathbf{v}_g \cdot \boldsymbol{\omega}_g, \quad \chi_P = \boldsymbol{\omega}_g \cdot \nabla \Phi_g, \quad b_0 = \beta \chi_g, \quad E_{L,R} = E_0 \mp b_0 \quad (9)$$

Dimensional check: v0.1.2 normalizes the analytic lane before testing parity. The physical dimensional coupling scale is intentionally not supplied by E011 and remains unresolved.

Registered code-to-mathematics steps

1. **S1** [G2; equations E1]: P00 verifies exact E011 archive provenance and emits only anonymous topology IDs.
2. **S2** [G2; equations E1]: P01 builds robust standardized geometry features and deterministic target-free clusters.
3. **S3** [G3; equations E2]: P02 tests circulation-sign odd/even covariance; optional geometry-resolved filament probes require raw source SHA verification.
4. **S4** [G3; equations E3, E4]: P03 ranks all anonymous knot pairs by preregistered nonnegative geometry amplitudes, discrete relative sign, provider penalty and 2+1 cancellation score.
5. **S5** [G3; equations E5, E6]: P04 evaluates universal $2\pi/4\pi$ quaternion lift separately from knot-specific discrete frame holonomy.
6. **S6** [G3; equations E2, E6]: P05 keeps circulation reversal, curve orientation and spatial mirror as distinct involutions and requires independent support before a two-state operator can PASS.
7. **S7** [G3; equations E9]: P06 tests parity-even gravity controls against parity-odd background pseudoscalars and verifies the induced sign reversal of an analytic axial L/R splitting. It cannot PASS the physical-mechanism subgate without a dynamics-ready background field and independently derived coupling scale.
8. **S8** [G3; equations E7]: P07 attempts no closure unless a second independently measured U1-like observable exists; definitionally constructing Y from Q and T is forbidden.
9. **S9** [G8; equations E8, E4]: P08 evaluates additive 3-component candidates and separately checks source readiness of link/container controls.

7 Algorithm and phase order

P00 source intake; P01 static structure; P02 conjugation; P03 signed-response discovery; P04 monodromy; P05 involutions; P06 gravity-chiral selection; P07 independent-U1 closure; P08 composites. The phase order is advisory for evidence inheritance but nonblocking for execution.

8 Sources and provenance

The embedded E011 archive is pinned by SHA-256. Geometry files referenced by E011 may enter optional numerical lanes only after exact raw-file hash verification and do not count as independent physical replication.

9 Gate semantics

After G1, G2–G8 are siblings depending only on the common source gate. Thus one scientific FAIL does not prevent another question from being evaluated. G9 is reveal integrity only.

10 Numerics and error controls

Static feature scaling uses median/MAD. Circulation covariance uses FP64 filament probes with an 10^{-11} parity tolerance when source geometry is available. Pair discovery uses a frozen cancellation threshold of 0.08 and provider-spread threshold 0.25. P06 uses exact analytic parity identities with tolerance 10^{-14} ; physical background-field/coupling evidence is not supplied by E011 and therefore remains unresolved.

11 Backend authority

Python/NumPy FP64 is authoritative for v0.1.2. No C++/GPU result is promoted in this version. A later dynamics version may add native parity gates without changing this frozen protocol.

12 Blindness and reveal

Blind outputs use HMAC anonymous object IDs. Exact semantic family assignments and composite target mappings are committed under private/reveal.json and are attached only after canonical framework reveal verification.

13 Statistics and multiplicity

Cluster discovery is target-free. Pair discovery searches all eligible pairs over a small preregistered observable library and reports complete ranked multiplicity rather than treating the best pair as a population-significance claim.

14 Automatic results

Manifest note: the framework-generated “Output manifest SHA-256” below records the pre-report snapshot used at render time. The authoritative final archive state is the final OUTPUT_MANIFEST.json written during packaging.

Automatic run provenance

Field	Recorded value
Protocol bundle SHA-256	662d038c472cdbf2b7f5729b668b5031c933898fb8681a34e7867519262f57b7
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Output manifest SHA-256	169987fe706af6b91b0c9f3f00cf02fdc16f9739536c3fde93a8e044dcdb54ae

Gate ledger

Gate	Status	Question	Reason
G0	PASS	Are frozen protocol, framework pin and instance implementation byte-consistent?	Framework verified frozen protocol before pipeline entry; A059 additionally verifies its implementation manifest.
G1	PASS	Is the frozen E011 static-ready atlas source present and provenance-qualified?	Embedded E011 archive must match its preregistered SHA-256 exactly.
G2	PASS	Do source-intake and static-structure numerics pass their preregistered sanity criteria?	P00 source intake and P01 static structure both PASS.
G3	PASS	Does target-free multi-phase discovery produce at least one nontrivial operator candidate while preserving every lane status separately?	At least one target-free operator lane PASSES; non-PASS sibling lanes remain explicitly preserved and prevent integrated mechanism closure.
G4	NOT_APPLICABLE	Is there an independent held-out physical confirmation source?	No independent held-out physical confirmation source is claimed in v0.1.2.
G5	NOT_APPLICABLE	Is a native C++ certification lane enabled for this version?	No native C++ certification lane is enabled in v0.1.2.
G6	NOT_APPLICABLE	Is an optional GPU screening lane enabled and parity-qualified?	GPU screening is disabled in v0.1.2.
G7	NOT_APPLICABLE	Is there an independent source-family replication capable of closing the cross-source gate?	E011 provider multiplicity is provenance structure, not an independent physical replication.
G8	UNRESOLVED	Does the integrated operator/composite program close all required mechanism sectors rather than only isolated diagnostics?	Integrated mechanism cannot close while any required sector remains UNRESOLVED/DEFERRED; explicit FAILs remain preserved in metrics.

G9	PASS	After blind finalization, do reveal commitments verify without changing any blind phase result?	Reveal prerequisites, blind-output integrity, and both nonced commitments verified.
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Backend manifest

```
{
  "authority": "PYTHON_NUMPY_FP64_STATIC_AND_DIAGNOSTIC",
  "framework_version": "1.0.6",
  "implementation": {
    "file_count": 32,
    "manifest_sha256": "e427fd70d7132205d31c7c3653b4334a681568e18d6756aebddf040780f8c708",
    "mismatches": [],
    "pass": true
  },
  "note": "No native/GPU lane is promoted in v0.1.2. Phase-specific evidence classes define the allowed  
↪ scientific interpretation.",
  "schema": "A059-BACKEND-MANIFEST-1"
}
```

Source manifest

```
{
  "manifest_sha256": "a31f1795c6f9404b0c16a17f844ced6e48505d20a456547d357ba69b1ef3c7a7",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "E011",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family": "E011_SKLSA_STATIC_READY_V030",
      "path":
        ↪ "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A059_SST_Quantum_Number_Roset ]
        ↪ ta_Mega_Falsifier\\A059-v0.1.2\\inputs\\E011_SKLSA_Static_Ready_Seed_Atlas_v0.3.0-outputs.zip",
      "provenance_family": "E010_E011_STATIC_GEOMETRY_CHAIN",
      "registry_hint": {
        "catalog_id": "E011",
        "hits": []
      },
      "role": "primary static geometry/provenance atlas",
      "sha256": "97cd1acbf23c102e4564f2d74fd329a2d03f32a29b8bd376ffa8b806c14246cc",
      "source_id": "E011_STATIC_ATLAS",
      "version": "v0.3.0"
    }
  ],
  "workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}
```

A059 v0.1.2 post-run discussion

Reveal-only semantic discussion is generated after commitment verification. Blind phase outcomes remain unchanged.

15 Allowed interpretation

A phase PASS certifies only its named finite-model invariant or covariance. In P06 an analytic parity check cannot by itself certify a physical gravity origin of chiral interactions. Universal rotation identities, fitted closures, static cancellation, and source-topology availability are not interchangeable with physical quantum numbers.

16 Reproducibility

Run the numbered command files in order. Each phase writes its own RESULT.json and phase manifest. Then run the blind umbrella finalizer, canonical reveal-if-allowed, and post-reveal scorecard command.